

RTL to GDSII using OpenLane (Sky130)

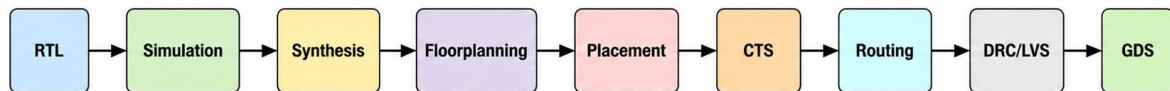
INTRODUCTION

This project demonstrates the complete **ASIC physical design flow** of an simple APB (Advanced Peripheral Bus) Controller from RTL to GDSII using the OpenLane framework with the Sky130 PDK.

The flow verifies the design functionality, converts RTL into a gate-level netlist, performs physical implementation, and generates a manufacturable layout.

RTL TO GDSII FLOW

RTL to GDSII flow is the process of converting a design written in Verilog (RTL) into a physical chip layout (GDS file) that can be manufactured.



RTL (Register Transfer Level)

RTL is the design stage where the digital circuit is described using a hardware description language like Verilog. It defines how data moves between registers and how logic operations are performed using constructs like always blocks and assign statements.

```

GNU nano 6.2 apb.v
module apb(
input pclk,
input presetn,

input [1:0] psel,
input pwrite,
input penable,

input [3:0] pwrdata,
input [3:0] paddr,

input [3:0] prdata0,
input [3:0] prdata1,
input [3:0] prdata2,

output reg [3:0] data,
output reg pready
);

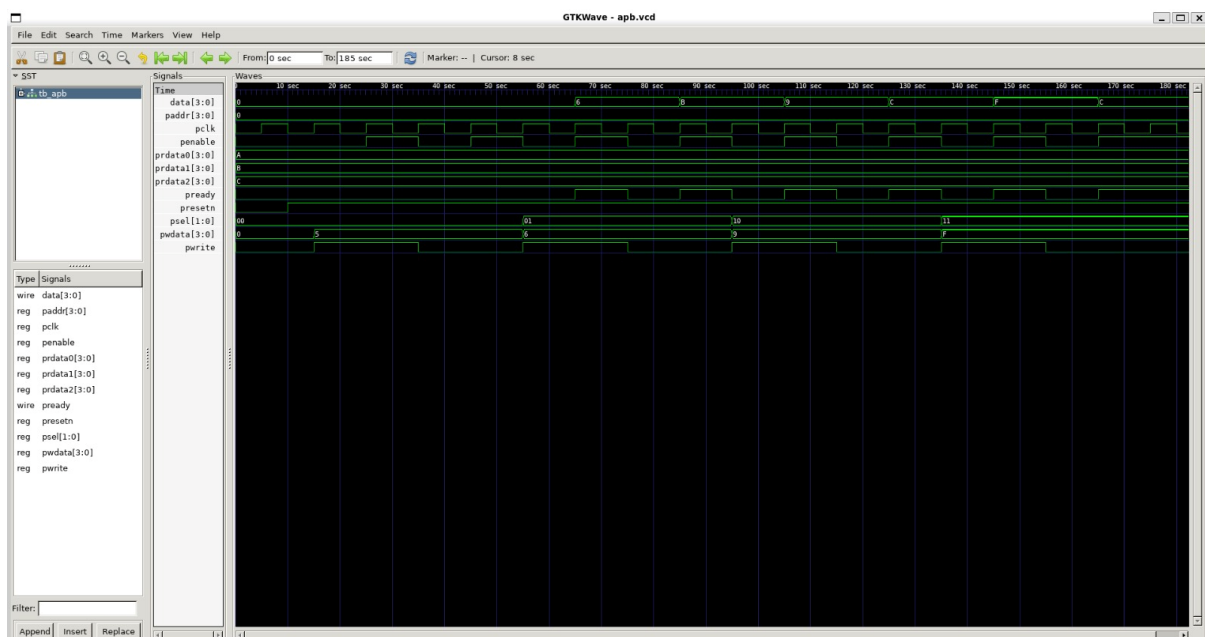
always @(posedge pclk or negedge presetn)
begin
if (!presetn)
begin
data <= 0;
pready <= 0;
end
else
begin
if (!psel && !penable)
begin
pready <= 0;
end
else if (!psel && penable)
begin
case (psel)
2'b00: begin
if (pwrite)

```

At this level, the design focuses on functionality and behavior, without any information about physical implementation or gates.

Simulation

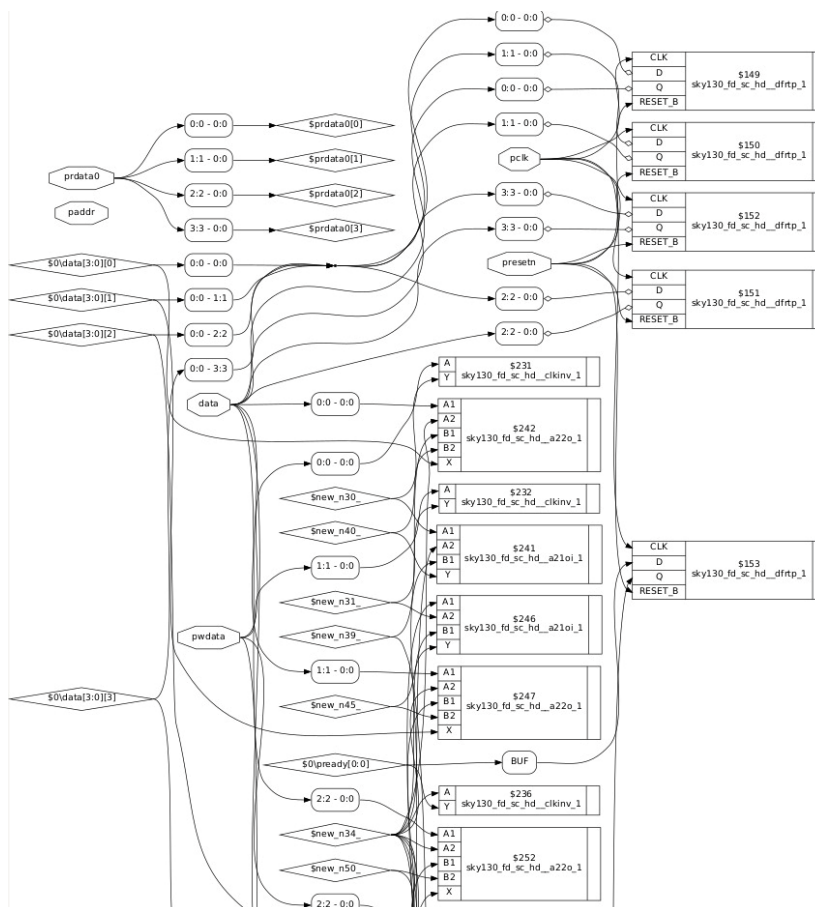
Simulation is used to verify that the RTL design works correctly before moving to synthesis. A testbench applies inputs such as clock, reset, and data to the design, and the outputs are observed through waveforms.



This helps detect logical errors early and ensures that the design behaves as expected under different

Synthesis

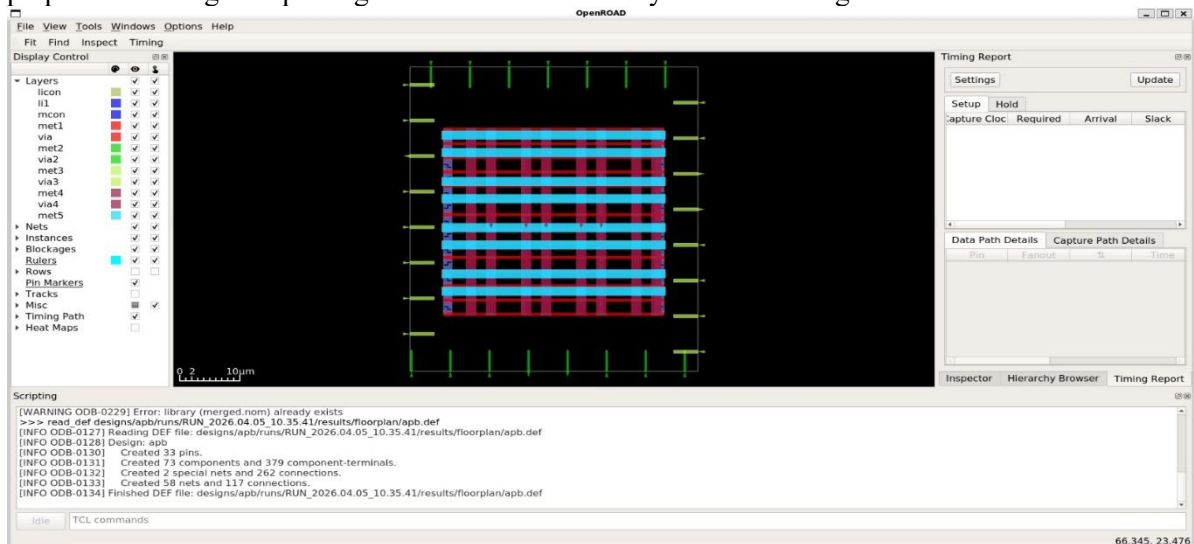
Synthesis converts the RTL design into a gate-level netlist using standard cells like AND, OR, and flip-flops.



It optimizes the design for timing, area, and power based on the given constraints. The output is a netlist (e.g., mapped.v) that represents the logical structure of the circuit using real hardware components.

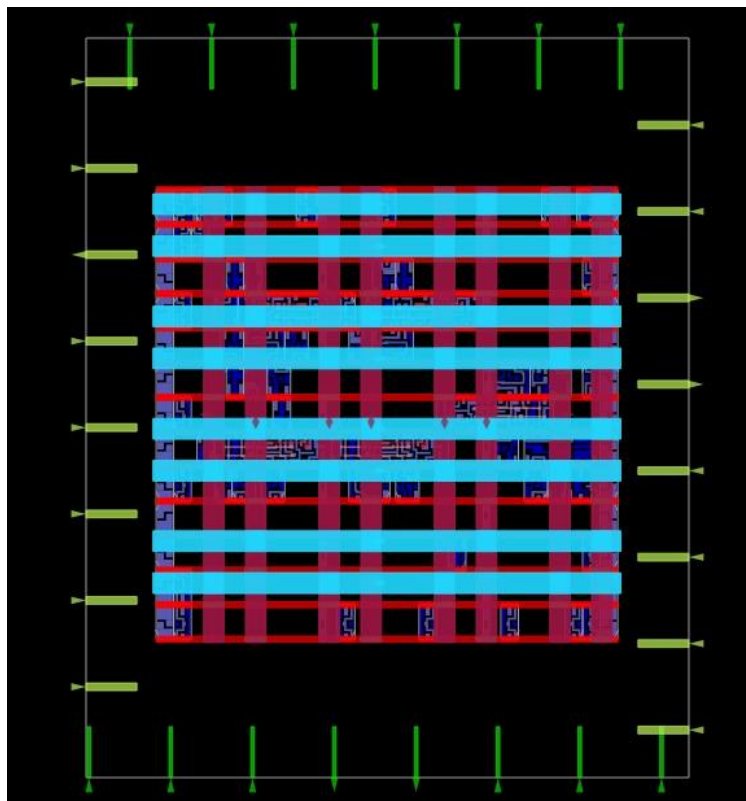
Floorplanning

Floorplanning defines the physical structure of the chip by deciding the chip size, core area, placement of input/output pins, and power distribution network. It sets up the layout boundaries and prepares the design for placing standard cells efficiently in the next stage.



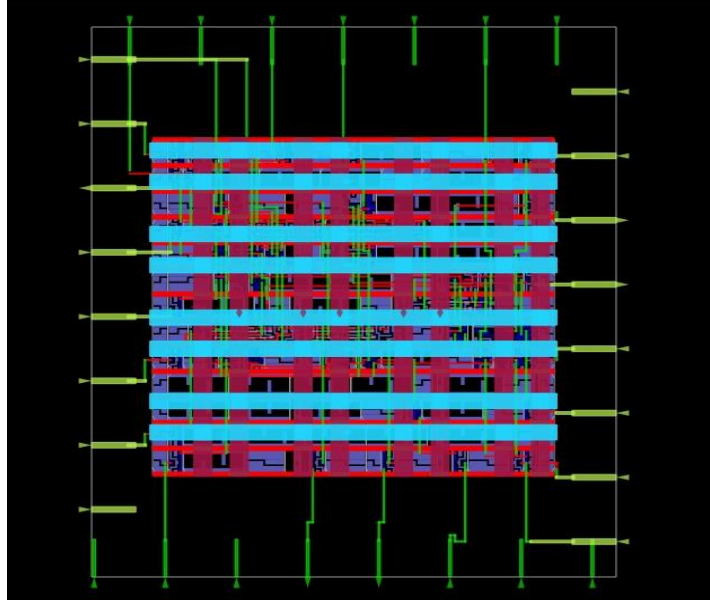
Placement

Placement arranges all the standard cells within the defined floorplan area. The goal is to minimize wire length, reduce delay, and avoid congestion while maintaining proper connectivity. At this stage, cells are placed but not yet connected with metal wires.



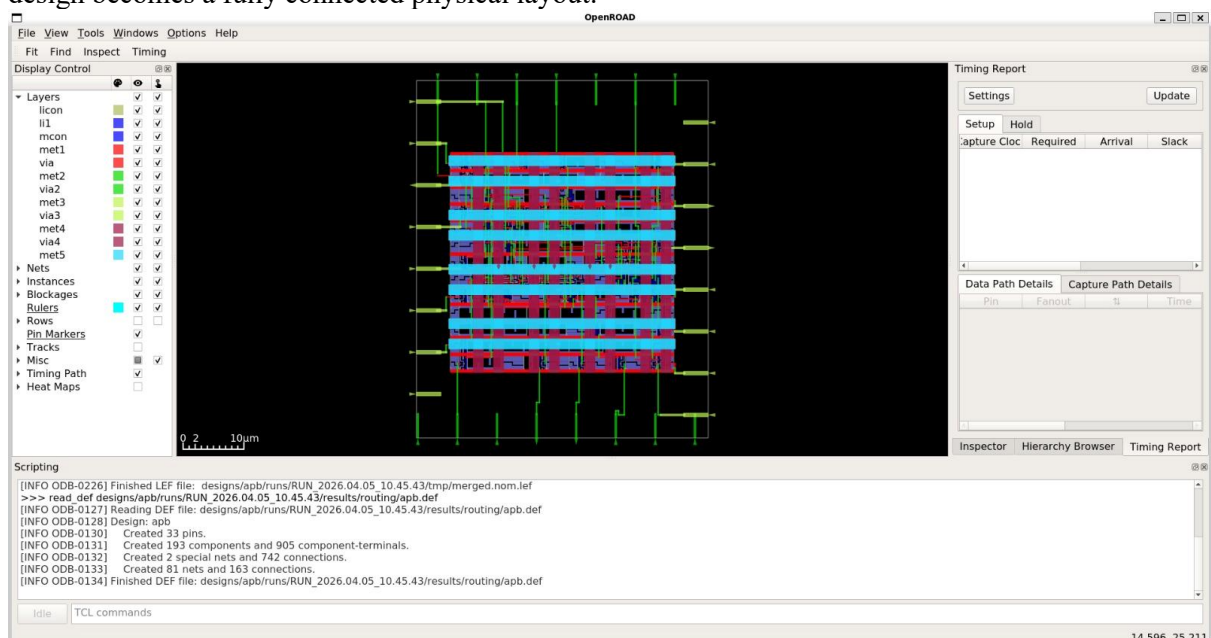
Clock Tree Synthesis (CTS)

CTS builds a clock distribution network that delivers the clock signal to all flip-flops in the design. Buffers are inserted to balance the clock path and reduce skew, ensuring that all sequential elements receive the clock signal at nearly the same time for correct operation.



Routing

Routing connects all the placed cells using metal layers to form complete electrical connections. It includes global routing (planning paths) and detailed routing (final connections). After this step, the design becomes a fully connected physical layout.



DRC (Design Rule Check)

DRC checks whether the layout follows manufacturing rules such as minimum spacing, width, and layer constraints. It ensures that the design can be fabricated without issues. A result of zero errors indicates a clean and manufacturable layout.

```

[STEP 40]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/routing/40-grt_sta.log)...
[STEP 41]
[INFO]: Running Fill Insertion (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/routing/41-fill.log)...
[STEP 42]
[INFO]: Running Detailed Routing (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/routing/42-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 43]
[INFO]: Checking Wire Lengths (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/routing/43-wire_lengths.log)...
1777802478
% run_magic
[STEP 44]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/signoff/44-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/apb/runs/RUN_2026.05.03_09.51.56/logs/signoff/44-lef.log)...
% run_magic_drc
[STEP 45]
[INFO]: Running Magic DRC (log: designs/apb/runs/RUN_2026.05.03_09.51.56/logs/signoff/45-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.

```

LVS (Layout vs Schematic)

LVS verifies that the physical layout matches the original logical design. It compares the extracted netlist from the layout with the synthesized netlist to ensure correct connectivity. A successful result confirms that the layout is logically accurate.

```

Circuit contains 75 nets, and 8 disconnected pins.
Contents of circuit 2: Circuit: 'apb'
Circuit apb contains 63 device instances.
  Class: sky130_fd_sc_hd__buf_1 instances: 18
  Class: sky130_fd_sc_hd__buf_2 instances: 4
  Class: sky130_fd_sc_hd__buf_4 instances: 1
  Class: sky130_ef_sc_hd__decap_12 instances: 1
  Class: sky130_fd_sc_hd__dfxtp_1 instances: 5
  Class: sky130_fd_sc_hd__clkinv_1 instances: 5
  Class: sky130_fd_sc_hd__decap_3 instances: 1
  Class: sky130_fd_sc_hd__decap_4 instances: 1
  Class: sky130_fd_sc_hd__decap_6 instances: 1
  Class: sky130_fd_sc_hd__decap_8 instances: 1
  Class: sky130_fd_sc_hd__nand2b_2 instances: 1
  Class: sky130_fd_sc_hd__a22o_1 instances: 4
  Class: sky130_fd_sc_hd__o21ai_0 instances: 8
  Class: sky130_fd_sc_hd__tapvpwrvrgnd_1 instances: 1
  Class: sky130_fd_sc_hd__o21ai_4 instances: 1
  Class: sky130_fd_sc_hd__a21oi_1 instances: 4
  Class: sky130_fd_sc_hd__fill_1 instances: 1
  Class: sky130_fd_sc_hd__fill_2 instances: 1
  Class: sky130_fd_sc_hd__nor2_1 instances: 4
Circuit contains 75 nets, and 8 disconnected pins.

Circuit 1 contains 63 devices, Circuit 2 contains 63 devices.
Circuit 1 contains 75 nets, Circuit 2 contains 75 nets.

Final result:
Circuits match uniquely.
.
Logging to file "/openlane/designs/apb/runs/RUN_2026.05.03_09.51.56/logs/signoff/52-apb.lef.lvs.log" disabled
LVS Done.

% |

```


GDSII Generation

GDSII generation produces the final layout file (.gds) that contains all geometric and layer information of the chip. This file is sent to the semiconductor foundry for fabrication, making it the final output of the entire design flow.

